Mathematical Problems in Deep Learning: Final Solutions

1. **Approximate DP:** Consider the following mechanisms M that takes a dataset $x \in [0,1]^n$ and returns an estimate of the mean $\bar{x} = (\sum_{i=1}^n x_i)/n$.

   (M1) $M_1(x) = \bar{x} + [Z]_{-1}^{1}$, for $Z \sim \text{Lap}(2/n)$.

   (M2)
   $$M_2(x) = \begin{cases} 1 & \text{w.p. } \overline{x} \\ 0 & \text{w.p. } 1 - \overline{x}. \end{cases}.$$

   In HW3, we have identified the above mechanisms that do not meet the definition of $(\epsilon, 0)$-differential privacy. For those mechanisms, calculate the smallest value of δ (again possibly as a function of n) for which they satisfy (ϵ, δ) differential privacy for a finite value of ϵ.

   Solution.

   (a) Suppose $\bar{x} = 1/n$ and $\bar{x'} = 0$. For any a, we have
   $$Pr(a \leq 1/n + [Z]_{-1}^{1} \leq 1 + 1/n) \leq e^{\epsilon} Pr(a \leq [Z]_{-1}^{1} \leq 1 + 1/n) + \delta.$$
   If $a > 1$, we have
   $$Pr(a - 1/n \leq [Z]_{-1}^{1} \leq 1) \leq \delta$$
   which implies
   $$\begin{aligned} \delta =& Pr(1 - 1/n \leq [Z]_{-1}^{1} \leq 1) \\ =& Pr(1 - 1/n \leq Z) \\ =& \frac{1}{2} e^{-(n/2)(1-1/n)} \\ =& \frac{1}{2} e^{-n/2+1/2}. \end{aligned}$$
   On the other hand, if $a < 1$, we have
   $$\begin{aligned} & Pr(a - 1/n \leq [Z]_{-1}^{1} \leq 1) \leq e^{\epsilon} Pr(a \leq [Z]_{-1}^{1} \leq 1 + 1/n) + \delta \\ \Leftrightarrow & Pr(a - 1/n \leq [Z]_{-1}^{1} \leq 1 - 1/n) \leq e^{\epsilon} Pr(a \leq [Z]_{-1}^{1} \leq 1) \end{aligned}$$
   which is true for $\epsilon = 1/2$. Thus, the above mechanism is $(1/2, 1/2e^{-n+1/2})$-DP. Note that you should specify ϵ as well since the mechanism is not (ϵ, δ)-DP for $\epsilon < 1/2$.

   (b)
   $$Pr(M_2(x) = 1) \leq e^{\epsilon} Pr(M_2(x') = 1) + \delta \Leftrightarrow \bar{x} \leq e^{\epsilon} \bar{x'} + \delta.$$
   Thus, the δ should be
   $$\delta = \max_{x, x'} \bar{x} - e^{\epsilon} \bar{x'} = 1/n.$$
   Thus, the above mechanism is $(0, 1/n)$-DP. Note that you should clearly mention that the mechanism is (ϵ, δ)-DP for all ϵ.

2. **Regression:** Consider a dataset where each of its n rows is a pair of real numbers (x_i, y_i), each from an interval $[-b, b]$. Suppose we wish to find a best-fit linear relationship $y_i \approx \beta x_i$ between the y's and the x's. Non-privately, a standard way to estimate β is via the OLS regression formula

$$\hat{\beta} = \hat{\beta}(x, y) = \frac{S_{xy}}{S_{xx}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

This is called *ordinary least-squares (OLS)* regression, since $\hat{\beta}$ is the minimizer of the mean-squared residuals

$$\frac{1}{n}\sum_i (y_i - \hat{\beta} x_i)^2. \tag{1}$$

(a) Show that the function $\hat{\beta}(x, y)$ has infinite global sensitivity, and hence we cannot get a useful DP estimate of it via a direct application of the Laplace or Gaussian mechanisms.

(b) Show that S_{xy} and S_{xx} have global sensitivity that is bounded solely as a function of b, and hence each of these can be approximated in a DP manner using the Laplace mechanism.

(c) Using Part 2b together with basic composition and post-processing, devise and implement an ϵ-DP algorithm for approximating $\hat{\beta}$ on an arbitrary dataset with $x_i, y_i \in \mathbb{R}$. In addition to the dataset $((x_1, y_1), \ldots, (x_n, y_n))$, your implementation should take as input parameters a clipping bound b and the privacy-loss parameter ϵ.

Solution.

(a) Let $X = \{(1, 0), \ldots, (0, 0)\}$ and $X' = \{(u, v), (0, 0), \ldots, (0, 0)\}$. Then, $\hat{\beta}(X) = 0$, and $\hat{\beta}(X') = v/u$. Since u can be arbitrarily close to 0 and the difference v/u is unbounded.

(b) Sensitivity of S_{xy} is $b \cdot b - b \cdot (-b) = 2b^2$ while sensitivity of S_{xx} is $b \cdot b - 0 \cdot 0 = b^2$.

(c) Based on composition theorem, you can mix two privatization with $\alpha\epsilon$ and $(1-\alpha)\epsilon$. In other words, compute $\tilde{S}_{xx}(X) = S_{xx}(X) + Z_{xx}$ where Z_{xx} be Laplace random variable with parameter $b^2/\alpha\epsilon$. Then, compute $\tilde{S}_{xy}(X) = S_{xy}(X) + Z_{xy}$ where Z_{xy} be a Laplace random variable with parameter $2b^2/(1-\alpha)$. Then, we compute $\tilde{\beta} = \tilde{S}_{xy}/\tilde{S}_{xx}$. The combining $\tilde{S}_{xx}$ and $\tilde{S}_{xy}$ is ϵ-DP due to composition theorem, and the final division does not affect the overall privacy due to post processing property.

Finally, you need to discuss the best α that minimizes the (any reasonable) loss between β and $\hat{\beta}$.

3. (a) Show that $\sigma(x) = \sin(x)$ is discriminatory.

(b) Show that $\sigma(x) = \sin(x) + \cos(x)$ is discriminatory.

Solution.

(a) Let μ be a measure, then

$$\int \sin(a^\intercal x) d\mu(x) = 0 \tag{2}$$

for all a. Also,

$$\begin{aligned} \int \sin(-a^\intercal x + \pi/2) d\mu(x) =& \int \cos(a^\intercal x) d\mu(x) && (3) \\ =& 0. && (4) \end{aligned}$$

Thus, the Fourier transform of μ is

$$\hat{\mu}(a) = \int e^{ia^\intercal x} d\mu(x) = 0 \tag{5}$$

for all a, which implies $\mu \equiv 0$. Thus, $\sin(x)$ is discriminatory.

(b) Let μ be a measure, then

$$\int \sin(a^\intercal x) + \cos(a^\intercal x) d\mu(x) = 0 \tag{6}$$

for all a. Also,

$$\begin{aligned} \int \sin(-a^\intercal x) + \cos(-a^\intercal x) d\mu(x) =& \int -\sin(a^\intercal x) + \cos(a^\intercal x) d\mu(x) && (7) \\ =& 0. && (8) \end{aligned}$$

Thus, we have

$$\int \sin(a^\intercal x) d\mu(x) = \int \cos(a^\intercal x) d\mu(x) = 0 \tag{9}$$

Finally,

$$\hat{\mu}(a) = \int e^{ia^\intercal x} d\mu(x) = 0 \tag{10}$$

for all a, which implies $\mu \equiv 0$. Thus, $\sin(x)$ is discriminatory.

4. Let $\Omega \subseteq \mathbb{R}^d$ be compact and $\sigma \in \mathcal{C}(\mathbb{R})$ is discriminatory function. Let

$$\mathcal{N}^{(1)}(\sigma, d, N) = \{A \cdot + b \,|\, A \in \mathbb{R}^{N \times d},\, b \in \mathbb{R}^N\}$$

and

$$\mathcal{N}^{(k+1)}(\sigma, d, N) = \{A\sigma(f(\cdot)) + b \,|\, N_{\text{prev}} \in \mathbb{N},\, A \in \mathbb{R}^{N \times N_{\text{prev}}},\, b \in \mathbb{R}^N,\, f \in \mathcal{N}^{(k)}(\sigma, d, N_{\text{prev}})\},$$

where σ applies elementwise, for $k = 1, 2, \ldots$.

(a) Briefly describe why $\mathcal{N}^{(2)}(\sigma, d, 1)$ is dense in $(C(\Omega), \|\cdot\|_\infty)$.
(Hint: We did this in our lecture).

(b) Show that $\mathcal{N}^{(2)}(\sigma, d, m)$ is dense in $(C(\Omega;\mathbb{R}^m), \|\cdot\|_\infty)$ for all $m \in \mathbb{N}$ where $C(\Omega;\mathbb{R}^m)$ is the space of bounded continuous function $u : \Omega \to \mathbb{R}^m$.

(c) Show that $\mathcal{N}^{(L)}(\sigma, d, 1)$ is dense in $(\mathcal{C}(\Omega), \|\cdot\|_\infty)$ for $L \in \{2, 3, \ldots\}$.

Solution.

(a) $\mathcal{N}^{(2)}(\sigma, d, 1) = \operatorname{span}(\{\sigma(a^\intercal x + b) : a \in \mathbb{R}^{1\times d}, b \in \mathbb{R}\}$, which we showed dense in $(C(\Omega), \|\cdot\|_\infty)$.

(b) Let $g = (g_1, \ldots, g_m) \in C(\Omega, \mathbb{R}^m)$. For any $\epsilon > 0$, there exists $h_1, \ldots, h_m \in \mathcal{N}^{(2)}(\sigma, d, 1)$ such that $\|g_i - h_i\|_\infty < \epsilon/m$. Suppose h_i has width N_i, then we can construct the 2-layer network h with width $\sum_{i=1}^m N_i$ where

$$h(x) = (h_1(x), h_2(x), \ldots, h_m(x)).$$

It is clearly $h \in \mathcal{N}^{(2)}(\sigma, d, m)$ and

$$\|h - g\|_\infty \leq \sum_{i=1}^m \|h_i - g_i\|_\infty < \epsilon. \tag{11}$$

Thus, $\mathcal{N}^{(2)}(\sigma, d, m)$ is dense in $(C(\Omega;\mathbb{R}^m), \|\cdot\|_\infty)$.

(c) We will show that $\mathcal{N}^{(L)}(\sigma, d, d)$ is dense for all $L \geq 2$ using induction. For $L \geq 3$, suppose $\mathcal{N}^{(L-1)}(\sigma, d, d)$ is dense. Let $g \in (C(\Omega;\mathbb{R}^d, \|\cdot\|_\infty)$ and $\epsilon > 0$. First, since $\mathcal{N}^{(2)}(\sigma, d, d)$ is dense, there exists $h_2 = A_2\sigma(A_1x + b_1) + b_1 \in \mathcal{N}^{(2)}(\sigma, d, d)$ such that

$$\|h_2 - g\|_\infty < \epsilon/2.$$

On the other hand, since $\mathcal{N}^{(L-1)}(\sigma, d, d)$ is dense, there exists $\tilde{h} \in \mathcal{N}^{(L-1)}(\sigma, d, d)$ such that

$$\|\tilde{h} - (A_1 \cdot + b_1)\|_\infty < \epsilon' \tag{12}$$

where $\epsilon' > 0$ will be specified later.
Let $h = A_2\sigma(\tilde{h}(x)) + b_2 \in \mathcal{N}^{(L)}(\sigma, d, d)$. Then,

$$\begin{aligned}\|h(x) - h_2(x)\| =& \|A_2\sigma(\tilde{h}(x)) + b_1 - (A_2\sigma(A_1x + b) + b_2)\| \\ =& \|A_2(\sigma(\tilde{h}(x)) - \sigma(A_1x + b))\|\end{aligned}$$

Since σ is continuous, we can set ϵ' small enough so that $\|h - h_2\|_\infty < \epsilon/2$. Finally,

$$\|h - g\|_\infty \leq \|h - h_2\|_\infty + \|h_2 - g\|_\infty < \epsilon. \tag{13}$$